

OPC UA Server App

OPC UA Server for ctrlX OS 03VRS

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1 About this documentation

Editions of this documentation

Edition	Release date	Note
01	2024-10	First edition OPC UA Server app version 3.2
02	2025-01	OPC UA Server app version 3.4 New: ➔ Chapter 5.3.11 Lossless Subscription on page 25 Revision: ➔ Chapter 5.3.5 Certificate management on page 16: Description added

2 Important directions on use

2.1 Intended use

2.1.1 Introduction

Rexroth products are developed and manufactured to the state-of-the-art.

The products are tested prior to delivery to ensure operational safety and reliability.

⚠ WARNING

Personal injury and damage to property due to incorrect use of products!

The products may only be used as intended.

Failure to use the products as intended may cause situations resulting in property damage and personal injury.

NOTICE

Damages resulting from unintended use

Rexroth As the manufacturer does not assume any warranty, liability or compensatory claims for damages resulting from unintended use of the products. The user alone shall bear the risks of an unintended use of the products.

Before using Rexroth products, make sure that all the prerequisites for an intended use of the products are met:

- Personnel that in any way, shape or form uses Rexroth products must first read and understand the relevant safety instructions and be familiar with their intended use
- Leave hardware products in their original state, i.e., do not make any structural modifications. It is not permitted to decompile software products or alter source codes
- Do not install damaged or defective products or commission them
- It has to be ensured that the products have been installed as described in the relevant documentation

2.1.2 Areas of use and application

Products of the ctrlX series are suitable for Motion/Logic applications.

NOTICE

Products of the ctrlX series may only be used with the accessories, mounting parts, and other components specified in this documentation. Components that are not expressly mentioned must neither be attached nor connected. The same applies to cables and lines.

Only to be operated with the hardware component configurations and combinations expressly specified and with the software and firmware specified in the corresponding documentations and functional descriptions.

Products of the ctrlX series are suitable for single-axis as well as for multi-axis drive and control tasks. Device types with different equipment and interfaces are available for using the system in specific applications.

Typical areas of application:

- Building automation
- IoT and Security Gateway or Device
- Handling & Robotic

Controls of the ctrlX CORE series may only be operated under the mounting and installation conditions, in the position of normal use and under the ambient conditions (temperature, degree of protection, humidity, EMC, etc.) specified in the related documentations.

2.2 Unintended use

"Unintended use" refers to using the ctrlX products outside of the above-mentioned areas of application or under operating conditions and technical data other than described and specified in the documentation.

ctrlX products must not be used if they are exposed to following conditions:

- Operating conditions that do not meet the specified ambient conditions.
Operation under water, under extreme temperature fluctuations or under extreme maximum temperatures is prohibited
- Applications that have not been expressly authorized by Rexroth

3 Safety instructions

The Safety instructions contained in the available application documentation feature specific signal words (DANGER, WARNING, CAUTION or NOTICE) and, where required, a safety alert symbol (in accordance with ANSI Z535.6-2006).

The signal word is meant to draw the reader's attention to the safety instruction and identifies the hazard severity.

The safety alert symbol (a triangle with an exclamation point), which precedes the signal words DANGER, WARNING and CAUTION, is used to alert the reader to personal injury hazards.

The Safety instructions in this documentation are designed as follows:

▲ DANGER

In case of non-compliance with this safety instruction, death or serious injury **will** occur.

▲ WARNING

In case of non-compliance with this safety instruction, death or serious injury **could** occur.

▲ CAUTION

In case of non-compliance with this safety instruction, minor or moderate injury could occur.

NOTICE

In case of non-compliance with this safety instruction, property damage could occur.

4 Introduction into the OPC Unified Architecture

4.1 General information

OPC UA is the further development of the OPC industrial standard. The service-oriented architecture ensures platform independency, scalability and high-availability by omitting a DCOM basis. OPC UA ensures a complete vertical integration from the control level up to the automation component irrespective of the programming language or the operating system. OPC UA is a client/server system. Several clients can access a server simultaneously. A client can access several servers.

4.2 Overview on specifications

OPC Unified Architecture was published as multi-part specification in 14 parts by the OPC foundation → <http://www.opcfoundation.org>. Only registered users can download the specification. Register on → <http://www.opcfoundation.org>.

Table 1: OPC UA - Overview on specifications

Part	Specification	Download
Part 1	OPC UA Part 1 - Overview and Concepts	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-1-overview-and-concepts/
Part 2	OPC UA Part 2 - Security Model	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-2-security-model/
Part 3	OPC UA Part 3 - Address Space Model	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-3-address-space-model/
Part 4	OPC UA Part 4 - Services	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-4-services/
Part 5	OPC UA Part 5 - Information Model	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-5-information-model/
Part 6	OPC UA Part 6 - Mappings	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-6-mappings/
Part 7	OPC UA Part 7 - Profiles	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-7-profiles/
Part 8	OPC UA Part 8 - Data Access	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-8-data-access/
Part 9	OPC UA Part 9 - Alarms and Conditions	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-9-alarms-and-conditions/
Part 10	OPC UA Part 10 - Programs	→ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-10-programs/

Part	Specification	Download
Part 11	OPC UA Part 11 - Historical Access	➔ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-11-historical-access/
Part 12	OPC UA Part 12 - Discovery	➔ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-12-discovery/
Part 13	OPC UA Part 13 - Aggregates	➔ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-13-aggregates/
Part 14	OPC UA Part 14 - PubSub	➔ http://www.opcfoundation.org/developer-tools/specifications-unified-architecture/part-14-pubsub/

4.3 Information model

The information model of the OPC UA is based on nodes describing an object-oriented context. A node can consist of attributes, methods and events. The content depends on the "NodeClass". The type model allows to map all types of user data including meta data. Each node has a unique "NodeId". References describe the connections between nodes. ReferenceType specifies the reference semantics.

The information model of the OPC UA is used as basis and extended as individual profile by other organizations such as PLCopen. For more information on these extensions, refer to the keyword "Companion Specification" or go to the respective organizations or the OPC foundation. Each OPC UA server reports the profiles it supports upon request.

4.4 Service-oriented architecture

The OPC UA architecture is divided into logical levels.

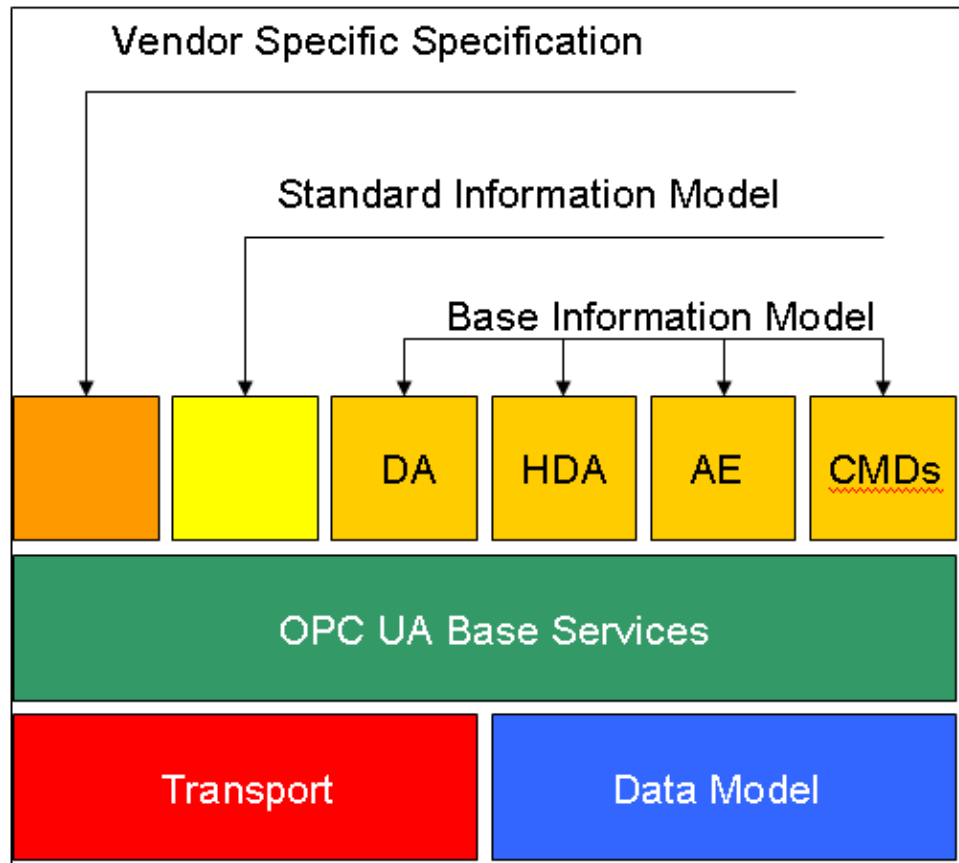


Fig. 1: OPC UA architecture

Transport level

The transport level serializes and deserializes data and sends or receives it. Data is transferred using an XML stream via the HTTP-HTTPS protocol or via a high-performance binary TCP protocol.

Basic OPC UA services

All basic OPC UA services are abstract method descriptions. They do not depend on the transport protocol and they are used as basis for all OPC UA functions. The methods are summarized in "Service Sets".

The most important "Service Sets" are mentioned in the following sections.

Discovery Service Set

The "Discovery Service Set" defines the services a client uses to determine the endpoints of a server and their security configuration. An endpoint is a variant consisting of a communication protocol, a host address, a port number and security settings.

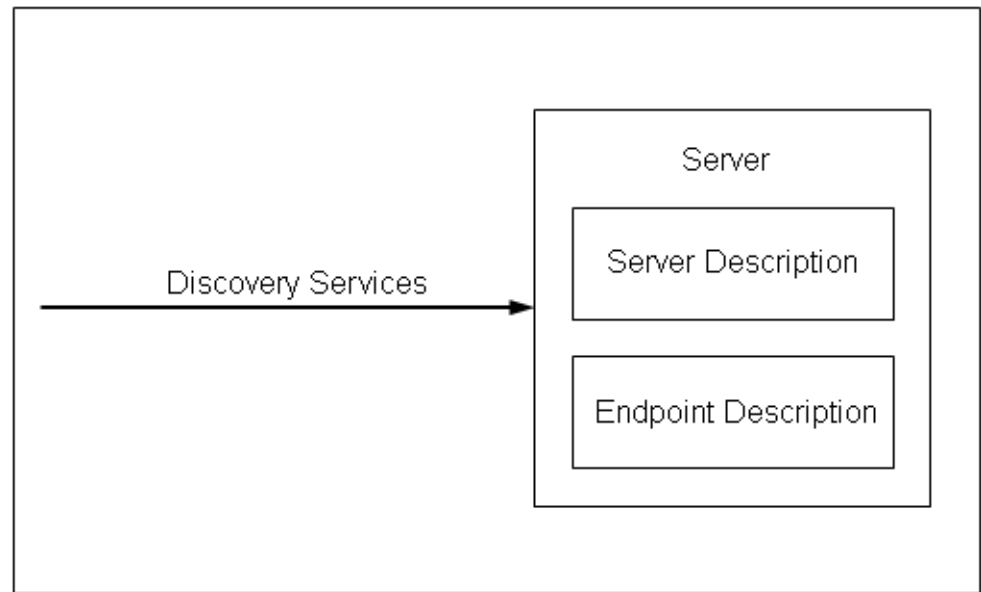


Fig. 2: Discovery Service Set

Session Service Set

A session is a logic connection between the client and the server on application level. The client can set up the connection. It can include client-specific user and language settings.

A session is set up using the "CreateSession" service. A session ID is assigned to the client. The client can use this session ID to call the following data services (Read, Write, Subscribe, etc.).

The server closes a session automatically after a client request (CloseSession) or after a timeout.

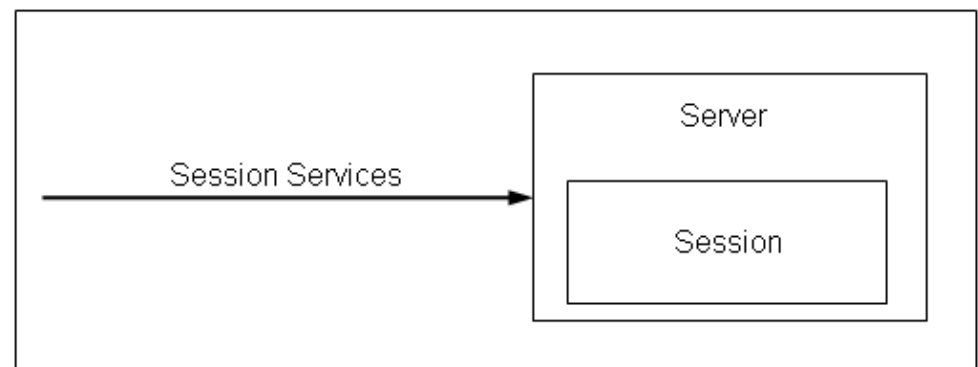


Fig. 3: Session Service Set

View Service Set

With the "View Service Set", the client allows the OPC UA server to browse through restricted address spaces (so-called views). The address space is intended to present available OPC UA server information to a client. It is generally generated from a number of nodes. The nodes are standardized objects and intended to map "Real World" objects.

Most important node classes:

- Variable
- Object
- Method

The connections between the nodes are set up using references. It is differentiated between hierarchical and non-hierarchical references. Hierarchical references are used to map a structural node order for example. Non-hierarchical references are used to refer to a node type definition for example.

The type system of nodes and references include types for objects, references, variables, events and data. This type system can be extended by the derivation of existing types and thus be provided with deeper semantics.

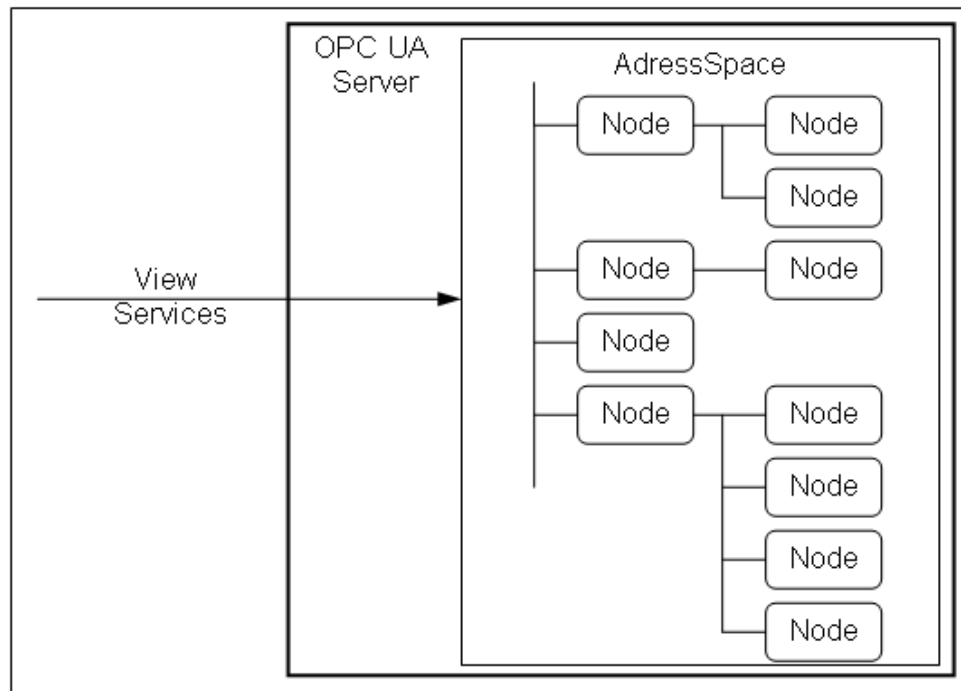


Fig. 4: View Service Set

Attribute Service Set

A node consists of multiple attributes. These attributes describe a node in detail. The node class specifies the attributes of a node. Only nodes of the "Variable" node class contain the attribute "Value".

All node classes have a common set of basic attributes. The basic attributes are for example "NodeId" to uniquely address a node, "NodeClass" and "BrowseName".

Read-only and write access is provided for these attributes using the "Service Set" attribute.

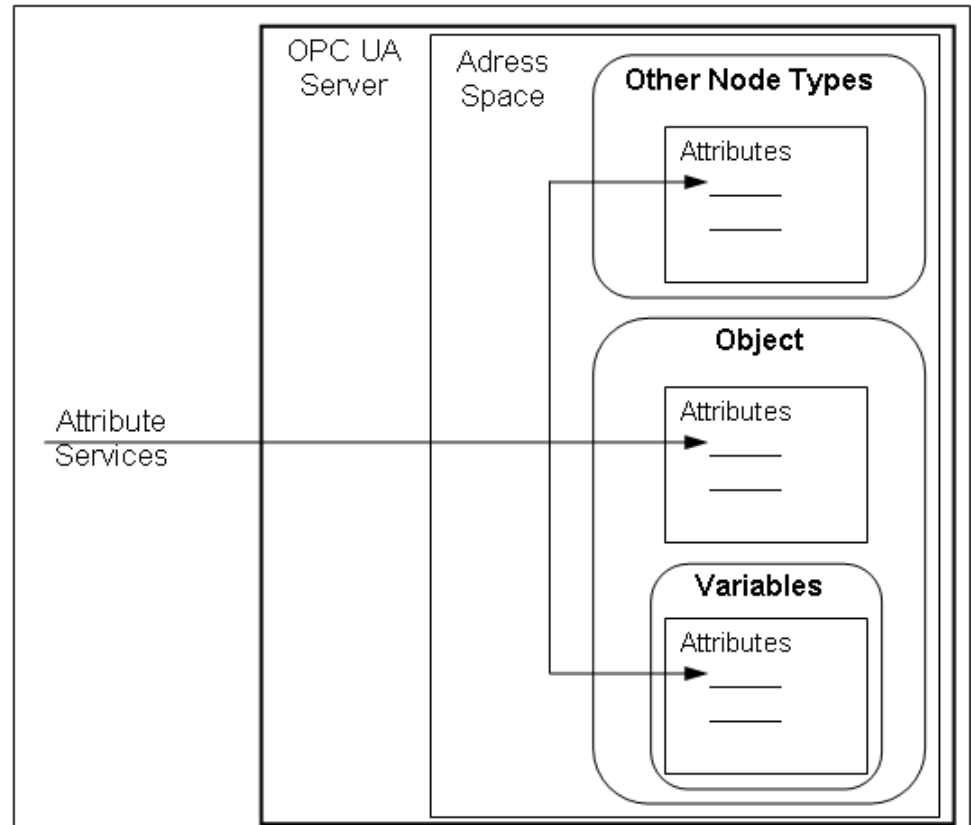


Fig. 5: Attribute Service Set

Subscription Service Set

Subscriptions are an efficient method in the OPC UA to receive information on value changes in the OPC UA server from the server. Clients can create, modify and delete subscriptions using the "Subscription Service Set"

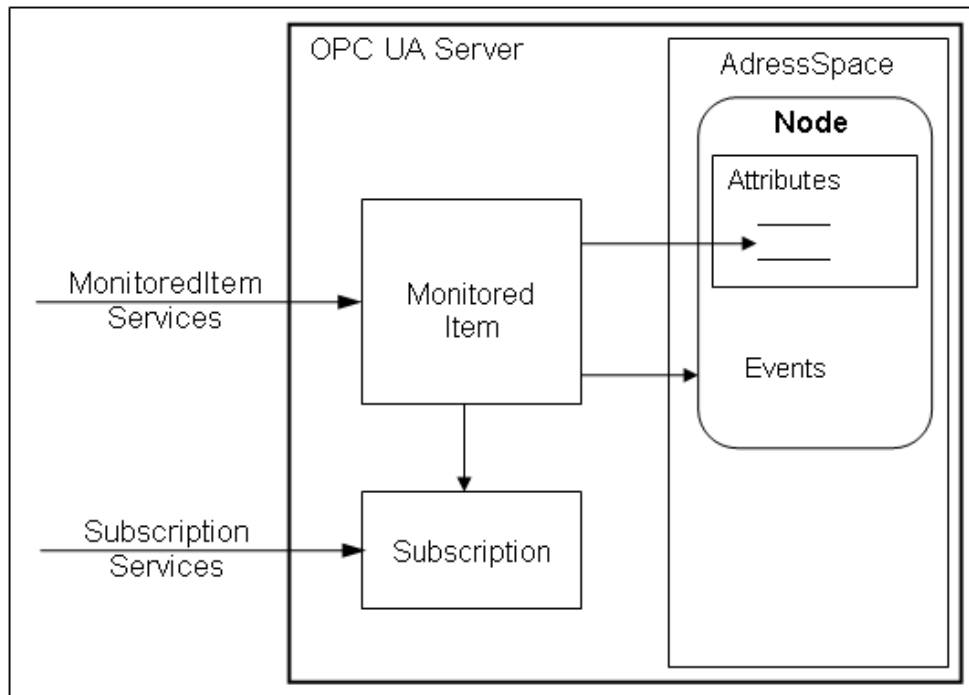


Fig. 6: Subscription Service Set

MonitoredItem Service Set

A "MonitoredItem" is a "Value" attribute, an aggregation of one or several "Value" attributes or an event included in a subscription. If the value of these "Value" attributes changes or if a corresponding event is reported, this value is transferred automatically to the client.

Clients can create, modify or delete "MonitoredItems" in the OPC UA server using the "MonitoredItem Service Set".

5 Rexroth ctrlX OPC UA Server

5.1 The ctrlX OPC UA Server on ctrlX OS

The ctrlX OPC UA Server runs as standalone app on the ctrlX OS.

Install the app on the ctrlX OS to add the standard communication protocol OPC UA to the ctrlX. This allows a standardized access to the device. The ctrlX backend Data Layer is used to access data of the ctrlX OS apps. This ensures that all data of the apps installed on the ctrlX OS is also available via the ctrlX OPC UA Server.

The "OPC UA Server" app is not installed on the ctrlX OS by default. Thus, install it on the ctrlX OS first.

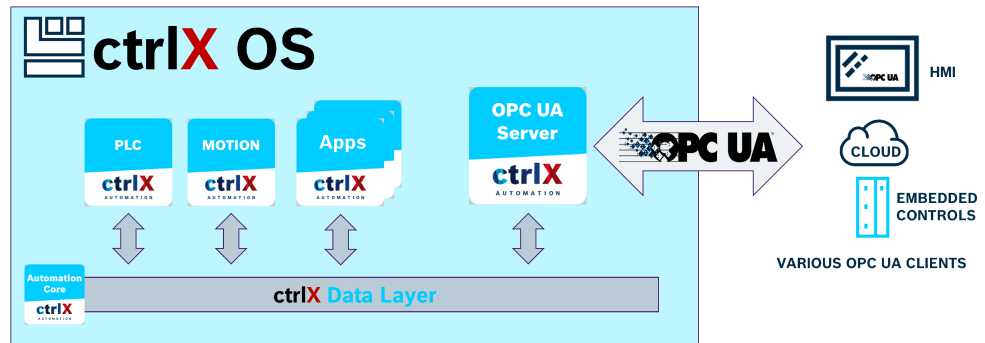


Fig. 7: ctrlX OPC UA Server in a ctrlX OS device

5.2 Installing on ctrlX OS

Install the app "OPC UA Server" on the ctrlX OS before using the ctrlX OPC UA Server. Use of the Package Manager of the ctrlX OS. The installation does not depend on other apps. All data ranges of the apps installed on the ctrlX OS are shown. The installation takes a few minutes. For a successful installation, the apps Device Admin and Automation Core have to be already installed on the ctrlX OS. It can only be installed in the "Service Mode". After the installation, return to the "Operation Mode" to provide control data in the ctrlX OPC UA Server.



To install on the ctrlX OS, the user needs the user permission "Apps: Manage apps" or "Administration: Full Access".

For the Package Manager, go to the web interface under "Settings → Apps".

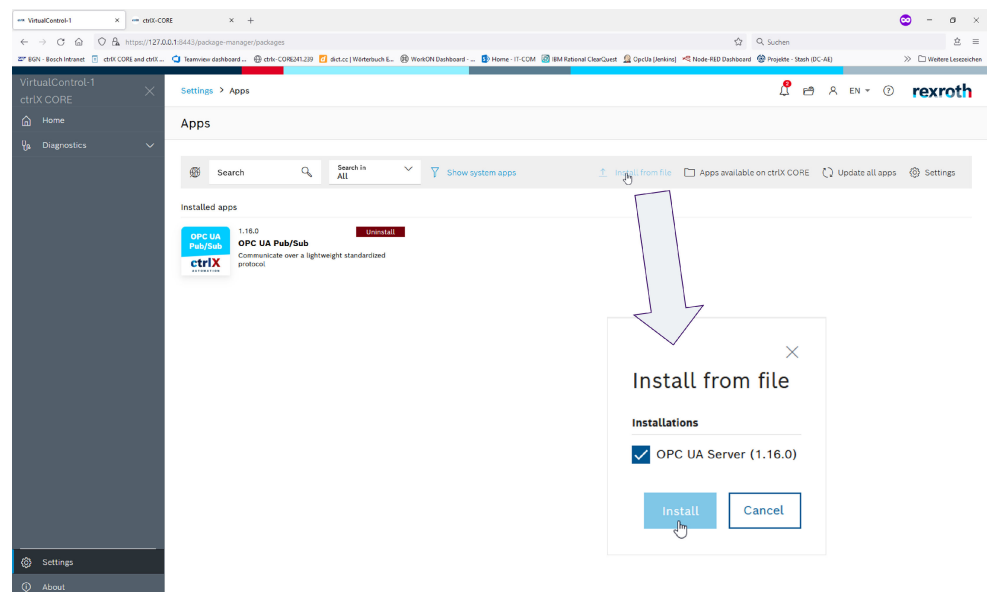


Fig. 8: For the installation of the app "OPC UA Server", go to "Settings → Apps". To install the app "OPC UA Server", use the web engineering interface.

To install, proceed with the following steps as shown in Fig. 8:

1. Select "Apps" from the "Settings" menu
2. Select to install the app from the file system
3. Select the installation file.
4. Install via the "Install" button.

After installing the app “OPC UA Server” and before using the app for the first time, set the respective permission in the Identity Manager under “*Settings → Users & Permissions → Permissions*”.

[Settings](#) > [Users & Permissions](#) > [Permissions](#) >

OPC UA Scopes : OPC UA Server access

[Authorized users](#)
[Authorized groups](#)

☒ **boschrexroth**

☐ **rexroot (system user)**

Save

Close

Fig. 9: Identity manager under “*Settings → Users & Permissions → Permissions*”

The user "boschrexroth" is already selected for permission.

All users with one of the following permissions can access the ctrlX OPC UA Server:

- "OPC UA Scopes: OPC UA Server access"
- "Administration: Full Access"

The permission can also be set when configuring a user.



5.2.1 Licensing

A valid license is required to use the OPC UA Server. This license can be loaded before or after the installation.

To operate the OPC UA Server, one of the following licenses is required:

- SWL_XCR_ENGINEERING_4H
- SWL-XC_x-UAS-OPCUASERVER_{xx}-NNNN



The OPC UA Server can be configured without license via the web interface of the control. However, it cannot be connected to the control in this state via the OPC UA.

Further information

- ctrlX OS Runtime 02VRS, Application Manual, chapter "[ctrlX licenses](#)" (DOK-XCORE*-XCR***V02**-APRS-EN-P)
- [ctrlX Automation website](#)

5.3 Properties

5.3.1 Connection settings

The ctrlX OPC UA Server can be accessed via all TCP/IP-capable network interfaces of the ctrlX OS device. With ctrlX CORE, the connections XF10, XF11, XF12 and XF51 are thus supported by the OPC UA Server.

The OPC UA Server uses the port 4840 by default.

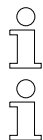
EndpointURL structure: **opc.tcp://<HostNameOfThe Control>:4840** or **opc.tcp://<HostAddress>:4840**.

The default IP address of the ctrlX OS device (192.168.1.1) corresponds to die EndpointURL **opc.tcp://192.168.1.1:4840**.

5.3.2 Security

The following methods to encrypt and sign (SecurityPolicies) are enabled by default:

- **SecurityPolicy [B] – Basic256Sha256**
- **SecurityPolicy [A] - Aes128-Sha256-RsaOaep**
- **SecurityPolicy - Aes256-Sha256-RsaPss**



Use the Aes256-Sha256-RsaPss method if possible, as it is the most secure method.

An unencrypted connection (SecurityPolicy None) can be enabled via the configuration.

An unencrypted connection is only recommended for diagnostic purposes.

The unencrypted connection should not be used for the general operation.

5.3.3 Endpoints

The following endpoints with different methods to encrypt and sign are available by default:

- **Sign –Basic256Sha256**
- **Sign – Aes128-Sha256-RsaOaep**
- **Sign – Aes256-Sha256-RsaPss**
- **SignAndEncrypt –Basic256Sha256**
- **SignAndEncrypt – Aes128-Sha256-RsaOaep**
- **SignAndEncrypt – Aes256-Sha256-RsaPss**

5.3.4 User und password

To connect to the ctrlX OPC UA Server, a valid combination consisting of username and password is required. Use the ctrlX OS Identity Manager to configure username and password (under “*Settings → Users & Permissions*” in the web interface).

The user has to have one of the following permissions:

- OPC UA Scopes: OPC UA Server access
- Administration : Full access

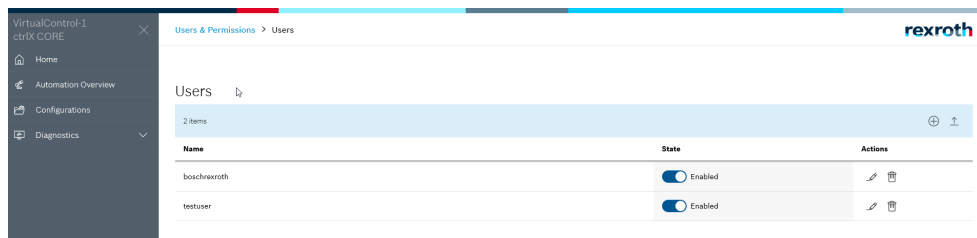


Fig. 10: Example configuration of a user in the ctrlX OS Identity Manager

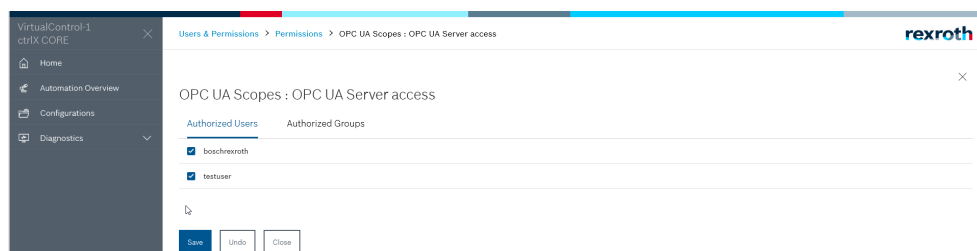
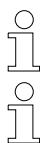


Fig. 11: Example configuration of the permissions in "OPC UA Scopes: OPC UA Server access"

To send the password to the control, the following two UsernameIdentityTokens are provided by default:

- "Username_256_Token" with SecurityPolicy BASIC256SHA256}
- "Username_256_RSAPSS_Token" with SecurityPolicy Aes256-Sha256-RsaPss



Username_256_Token with SecurityPolicy BASIC256SHA256 is also the standard method of SecurityPolicy Aes128-Sha256-RsaOaep.

The AnonymousIdentityToken "Anonymous_Token" is provided for the Find-Server & Get Endpoint services. Username and password are required to set up an OPC UA session. An anonymous access is not possible.

5.3.5 Certificate management

A secure connection between the OPC UA Server and the OPC UA Client is based on trustworthy certificates. Either self-signed certificates or certificates signed by a CA are currently supported here. The self-signed certificate management is based on a manual exchange of certificates between the OPC UA Server and Client. Certificates are exchanged when setting up a connection between the client and the server. Server and client have to trust the certificates manually. With CA-signed certificate management, the certificates are issued by a CA (Certificate Authority) and distributed to the corresponding servers and clients. The OPC UA Server and Clients only have to trust the CA certificate and manage the Certificate Revocation List (CRL) provided by the CA.

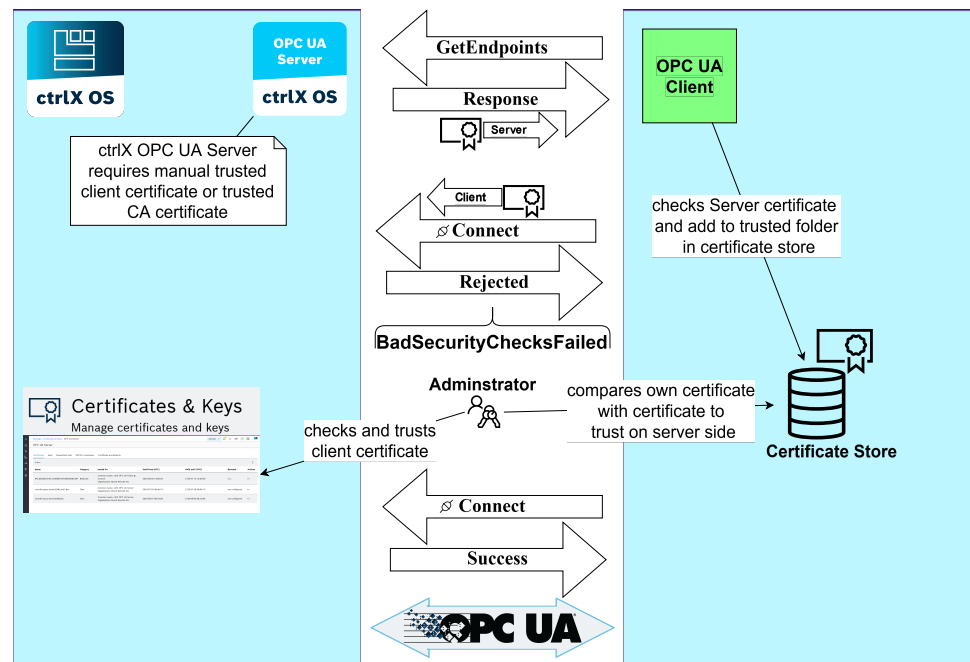


Fig. 12: Manual certificate management according to "OPC UA Specification Part 2: Security"

For the ctrlX OPC UA Server, the certificates are trusted via the ctrlX OS Certificate Manager (under "Settings → Certificates & Keys" in the web interface).

The following steps are required:

1. First, set up a secure connection (SecureChannel) using the OpenSecureChannel. The certificate is transferred from the client to the server.
 - ➔ The server reports the error "BadSecurityChecksFailed" and enters the client certificate into its "Reject" list for certificates.
2. The certificate of the client is shown as "rejected" in the Certificate Manager under "Certificates & Keys → OPC UA Server". This certificate can be trusted manually. The client has to trust the server certificate as well.
3. The client can now set up a secure OPC UA connection to the server.

The certificate of the client can also be uploaded directly in the Certificate Manager. Thus, step 1 is omitted. However, the certificate has to be renamed before. The file name has to correspond to the SHA1 value of the file. This can for example be determined using the fingerprint of the certificate. When renaming, all characters of the SHA1 have to be represented in capitals. The file schema is "[SHA1 value capitalized].der".

As of version 2.2, certificates can also loaded via the app data of the OPC UA Server. For this purpose, the certificates and any associated CRLs have to be stored in the "cert-import" folder in the OPC UA Server app data (path *opcuaserver*). The content of the folder "cert-import" is checked upon the start of the OPC UA Server app and any certificates included are checked and moved to the respective folders in the OPC UA Server Certificate Store.

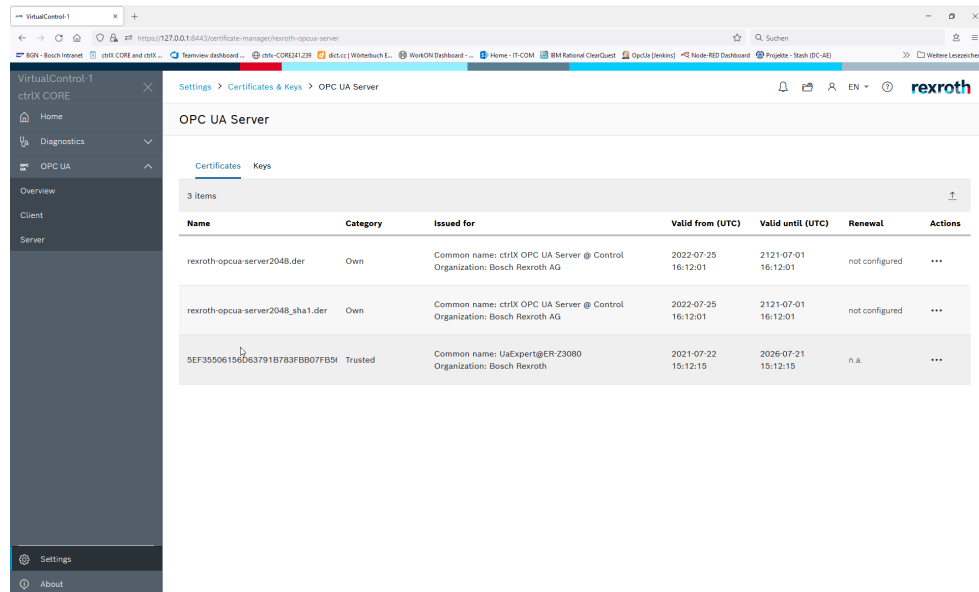


Fig. 13: Certificate Manager for the ctrlX OPC UA Server

Technical information on certificates and private keys

The certificates of the ctrlX OPC UA Server are currently self-generated and signed and valid for 36135 (ca. 99 year) by default.

The encryption methods used are "sha256" and "sha1".

The ctrlX OPC UA Server supports DER-encoded certificates and CRLs (Certificate Revocation List) according to the OPC UA specification. The private key has to be a PEM-encoded RSA private key.

Technical information on the storage of certificates

Certificates are stored in the Certificate Store.

For the Certificate Store, go to `$SNAP_COMMON/package-certificates/rexroth-opcua-server/rexroth-opcua-server/`.

The path for VirtualControl is for example:

`/var/snap/rexroth-opcua-server/common/package-certificates/rexroth-opcua-server/rexroth-opcua-server/`

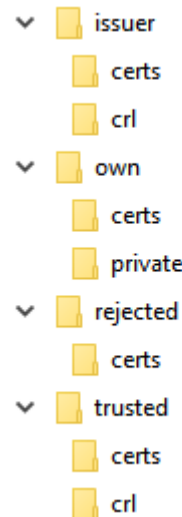


Fig. 14: Certificate Store - Folders and subfolders

The Certificate Store consists of the following folders:

- **issuer:**
Currently not supported
- **own:**
Includes the Application Instance Certificate from the OPC UA Server or Client and the respective private keys
- **rejected:**
Includes certificates from the UA Client or Server that intend to set up a connection to the OPC UA Server or Client, but they have not yet been trusted
- **trusted:**
Includes certificates of the UA Clients or Servers. The OPC UA Server or Client trusts these certificates
The respective subfolder “certs” includes the certificates belonging to the individual categories, e.g. which certificates can be trusted (below the folder “trusted”) and which cannot be trusted (below the folder “rejected”).
There is a Certificate Revocation List (CRL) in the subfolder “crl”.

5.3.6 Protocol and encoding

The following is supported as per the profile of the OPC UA standard release 1.04 (Part 7 - Profiles):

Table 2: ctrlX OPC UA Server - Supported profiles

Protocol and Encoding		
Name	Description	From Profile
Protocol UA TCP	Support the UA TCP transport protocol defined in UA Part 6.	UA-TCP UA-SC UA-Binary
UA Secure Conversation	Support UA Secure Conversation specified in UA Part 6.	UA-TCP UA-SC UA-Binary
UA Binary Encoding	Support UA Binary Encoding. Values of these data types are encoded in compact binary formats, contiguously and without tagging. I.e. the receiver is assumed to understand the structure it is decoding.	UA-TCP UA-SC UA-Binary

5.3.7 Supported services

The following services are supported as per the profile of the OPC UA standard release 1.04 (Part 7 - Profiles):

Table 3: ctrlX OPC UA Server - "Discovery Services"

Discovery Services		
Name	Description	From Profile
Discovery Get Endpoints	Support the GetEndpoints Service to obtain all Endpoints of the Server. This includes filtering based on Profiles.	Core 2017 Server Facet
Discovery Find Servers Self	Support the FindServers Service only for itself	Core 2017 Server Facet

Table 4: ctrlX OPC UA Server - "Session Services"

Session Services			
Name	Description	From Profile	Comments
Session General Service Behaviour	Implement basic Service behaviour. This includes in particular: - checking the authentication token - returning the requestHandle in responses - returning available diagnostic information as requested with the 'returnDiagnostics' parameter - respecting a timeoutHint	Core 2017 Server Facet	
Session Base	Support the Session Service Set (CreateSession, ActivateSession, CloseSession) except the use of ActivateSession to change the Session user. This includes correct handling of all parameters that are provided. Note that for the CreateSession and ActivateSession services, if the SecurityMode = None then: 1) The Application Certificate and Nonce are optional. 2) The signatures are null/empty. The details of this are described in Part 4.	Core 2017 Server Facet	
Session Minimum 1	Support minimum 1 Session (total)	Core 2017 Server Facet	Max. 50 Sessions Max. 10 Sessions for Subscriptions
Session Minimum 2 Parallel	Support minimum 2 parallel Sessions (total for all Clients)	Micro Embedded Device 2017 Server Profile	Max. 50 Sessions Max. 10 Sessions for Subscriptions

Table 5: ctrlX OPC UA Server - "View Services"

View Services			
Name	Description	From Profile	Comments
View Basic	Support the View Service Set (Browse, BrowseNext)	Core 2017 Server Facet	
View TranslateBrowsePath	Support Translate-BrowsePathsToNodeIds Service	Core 2017 Server Facet	

Table 6: ctrlX OPC UA Server - "Attribute Services"

Attribute Services		
Name	Description	From Profile
Attribute Read	Supports the Read Service to read one or more Attributes of one or more Nodes. This includes support of the IndexRange parameter to read a single element or a range of elements when the Attribute value is an array	Core 2017 Server Facet
Attribute Write Values	Supports writing to values to one or more Attributes of one or more Nodes	Core 2017 Server Facet

Table 7: ctrlX OPC UA Server- "Supscription Services"

Subscription Services			
Name	Description	From Profile	Comments
Subscription Basic	Support the following Subscription Services: CreateSubscription, ModifySubscription, DeleteSubscriptions, Publish, Republish and SetPublishingMode	Embedded Data-Change Subscription Server Facet	
Subscription Minimum 05	Support at least 5 Subscriptions per Session. This number has to be supported for at least half of the minimum required sessions.	Enhanced DataChange Subscription 2017 Server Facet	Max 150. Subscriptions per Session
Subscription Publish Min 10	Support at least 10 Publish Service requests per Session. This number has to be supported for at least half of the minimum required sessions. Support, as a minimum, the number of Publish requests per session as the size of the NotificationMessage retransmission queue for Republish.	Enhanced DataChange Subscription 2017 Server Facet	Max 10. Publish Service requests per Session

Subscription Services			
Name	Description	From Profile	Comments
Subscription Publish Discard Policy	Respect the specified policy for discarding Publish Service requests. If the maximum number of Publish Service requests has been queued and a new Publish Service request arrives, the "oldest" Publish request has to be discarded by returning the proper error.	Embedded Data-Change Subscription Server Facet	

Table 8: ctrlX OPC UA Server- "Monitored Item Services"

Monitored Item Services			
Name	Description	From Profile	Comments
Monitor Basic	Support the following MonitoredItem Services: CreateMonitoredItems, ModifyMonitoredItems, DeleteMonitoredItems and SetMonitoringMode	Embedded Data-Change Subscription Server Facet	
Monitor Value Change	Support creation of MonitoredItems for Attribute value changes. This includes support of the IndexRange to select a single element or a range of elements when the Attribute value is an array	Embedded Data-Change Subscription Server Facet	
Monitor Items 500	Support at least 500 MonitoredItems per Subscription. This number has to be supported for at least half of the required subscriptions for half of the required sessions.	Enhanced DataChange Subscription 2017 Server Facet	Max 500 pro Session. 5000 across all 10 Sessions
Monitor MinQueue-Size_05	Support at least 5 queue entries for MonitoredItems. Servers often will adapt the queue size to the number of currently MonitoredItems. However, it is expected that Servers support this minimum queue size for at least one third of the supported MonitoredItems.	Enhanced DataChange Subscription 2017 Server Facet	Max Queue size: 100

Table 9: ctrlX OPC UA Server- "Method Services"

Method Services				
Name	Description	From Profile	Test Cases	Comments
Method Call	Support the Call Service to call (invoke) a Method which includes support for Method Parameters.	Standard Data-Change Subscription 2017 Server Facet	Open	Only partially supported in the data range of the Data Layer

5.3.8 Address space of the ctrlX OPC UA Server

Currently, there are ten namespaces in the ctrlX OPC UA Server:

Table 10: ctrlX OPC UA Server - Namespaces

Name-space Index	Namespace URI
0	http://opcfoundation.org/UA/
1	urn:Control@Rexroth:ctrlX:AUTOMATION:Server
2	http://www.boschrexroth.com/OpcUa/Datalayer
3	http://www.boschrexroth.com/OpcUa/DatalayerTypes
4	http://www.boschrexroth.com/OpcUa/DatalayerEncoding
5	http://www.boschrexroth.com/OpcUa/DatalayerEventTypes
6	http://www.boschrexroth.com/OpcUa/DatalayerMethod
7	http://www.boschrexroth.com/OpcUa/DatalayerEnumeration
8	http://www.boschrexroth.com/OpcUa/DatalayerObjects
9	http://www.boschrexroth.com/OpcUa/DatalayerInputArguments
10	http://www.boschrexroth.com/OpcUa/DatalayerOutputArguments

Namespaces 2 to 10 provide data for the Data Layer, its types and encoding of structures in the Data Layer.



The namespace URI identifies a unique namespace as a resource within the OPC UA address space and should not be confused with the URL of a website.

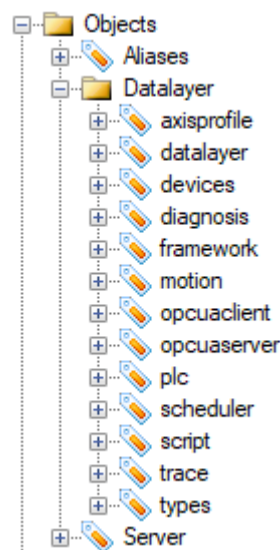


Fig. 15: ctrlX OPC UA Server - Address space under "Objects"

All payload of the control is below the node *Objects/DataLayer/*. Child data is directly mapped to the Data Layer. This area is adjusted dynamically depending on the installed snaps.

In the Fig. 15 for example, all data of the apps Automation Core, Motion, OPC UA Server and OPC UA Client is available.

5.3.9 Supported services for the data range Data Layer

Supported services for the Data Layer data range:

- **Browse und TranslateBrowsePathsToNodeIds**
- **Read**
- **Write**
- **Subscription**
- **Call**
- **Events**

5.3.10 OPC UA events

The ctrlX Data Layer provides an event mechanism. "Events" map events. This can be a machine error message for example. The Data Layer events can also be transmitted via the ctrlX OPC UA Server. With a OPC UA subscription to the Data Layer nodes, corresponding OPC UA events can be received.

The following figure exemplarily shows an OPC UA event subscription. Events are received from the Data Layer node *framework/bundles*. This node provides events when changing the app installations. In the example, the ctrlX Motion App was installed. This installation provides multiple events informing on the installed bundles.

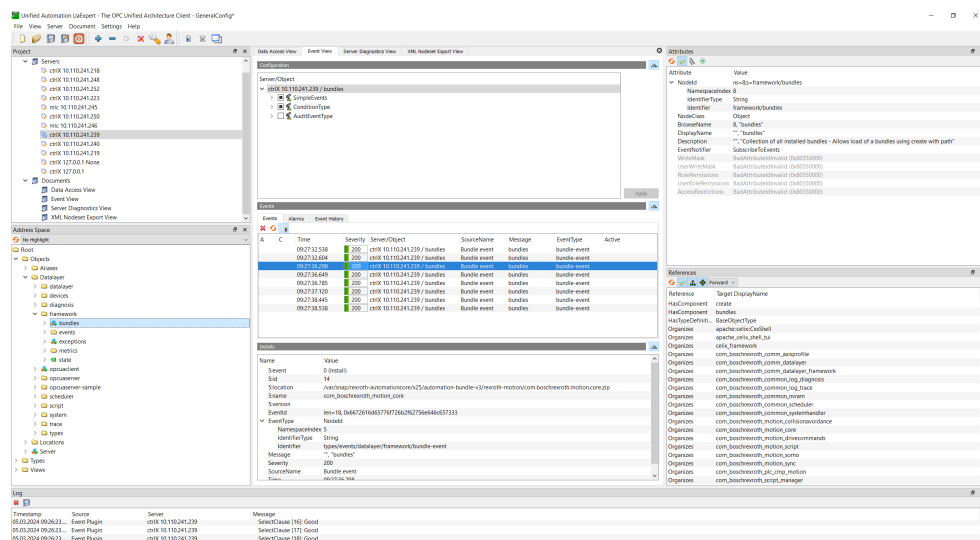


Fig. 16: Exemplary OPC UA event subscription

Current restrictions of the OPC UA events:

- When a node is added to the event subscription for the first time, this is acknowledged with an error. This is only carried out successfully the second time the node is added.
- The "EventNotifier" attribute initially returns the value "None" for all Data Layer nodes. The value "SubscribeToEvents" is only provided after the node has been added to the event subscription if the node can provide events.
- The OPC UA Client selects the event fields via "selectClause" defined as "SimpleAttributeOperand".

Only the following combination of "typeDefinitionId" and "browsePath" is currently supported:

- typeDefinitionId: ns=0;i=2041 (BaseEventType)
- browsePath: BrowseName of the event field, e.g.: 5, "name"

5.3.11 Lossless Subscription

In some applications, real-time data has to be subscribed so that all data changes of this node are provided. Thus, subscribe to the variable address below the RT base address. A Lossless Subscription ensures that all value changes to the RT variables are detected and transferred via the NRT subscription. In addition, some subscription properties can be configured.

This functionality is not available with a OPC UA Server with a basic license.

The sampling interval of the MonitoredItem has to be set to 0.

Use the Data Layer variable *opcuaserver/subscription-configuration/default-lossless-rate-limit* to set that only every nth value change is transferred.

Use the Data Layer variable *opcuaserver/subscription-configuration/default-lossless-interval* to set the time intervals in microseconds after which the value is transmitted. All value changes in the real-time range are detected, but only transferred after the interval has expired.

Use the Data Layer variable *opcuaserver/subscription-configuration/default-lossless-interval-tolerance* to set the tolerance interval in microseconds for *default-lossless-interval*.

Only either *default-lossless-rate-limit* or the combination of *default-lossless-interval* and *default-lossless-interval-tolerance* can be configured. Using both values is not possible. When the MonitoredItem is created, the global Data Layer configurations are retrieved and used for this MonitoredItem.

A possible OPC UA subscription configuration is shown in the following:

default-lossless-interval: 20000 microseconds

default-lossless-interval-tolerance: 1000 microseconds

OPC UA subscription

Publishing Interval: 200 milliseconds

OPC UA MonitoredItem

NodeId: plc/app/AppOpcUaServerRT/realtime_data/PRG_RTData-layer_GVL/data/GlobalCounter

Sampling Interval: 0

Queue Size: 20

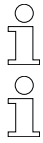
All value changes of the RT date are transferred when the interval of 20,000 microseconds expires. These are stored in the OPC UA subscription in the queue of the MonitoredItem and transferred after the 200 millisecond Publish-Interval has expired.

5.4 Configuration

The ctrlX OPC UA Server can be configured with the version 1.6.0 using the Data Layer. As shown in Fig. 17, all configuration parameters and configuration functions are below the Data Layer item "opcuaserver". A changed configuration is applied upon the next restart of the ctrlX Core OPC UA Server

The configuration also includes parameters for sessions, subscriptions, SecurityPolicies and endpoints.

In 1.12.0, basic parameters can also be configured via the web interface, see section [Chapter 5.5 Web interface on page 31](#).



An initial configuration is not required. The OPC UA Server app has a default configuration that allows an immediate server operation.

The OPC UA server might not be reachable anymore due to an incorrect configuration. In this case, either undo the change via the Data Layer or uninstall the ctrlX app OPC UA Server and install it again.

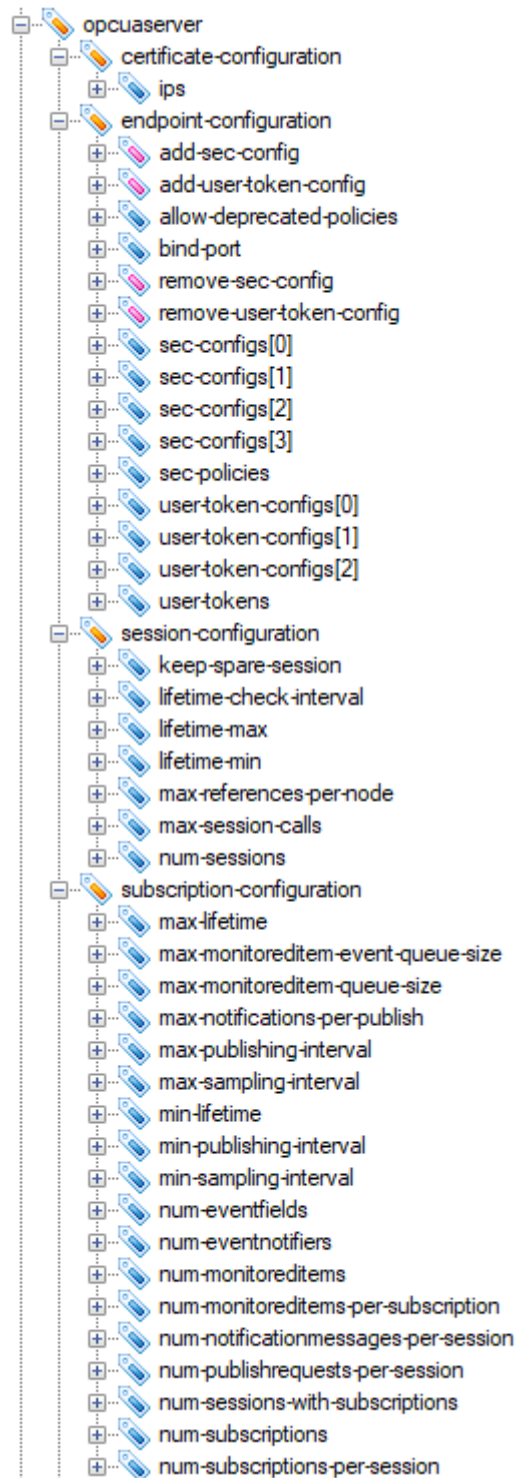


Fig. 17: Server configuration view in the address space of the ctrlX OPC UA Server

5.4.1 Certificate configuration

For the Certificate configuration, go to the Data Layer item *Certificate-configuration*.

The following configuration parameters (Data Layer items) are available:

Data Layer item	Data type	Description	Default value
<i>ips</i>	String	List of IPs separated by commas	

The *ips* field describes a list of IPs (e.g. "192.168.1.1,127.0.0.1") separated by a comma and written to the "IPAddresses" field when generating the certificate.



The certificate is only newly generated if the certificate (default:"rexroth-opcua-server2048.der") was deleted manually before via the ctrlX WebUI under "Settings → Certificate & Keys → OPC UA Server".

5.4.2 Endpoint configuration

For the endpoint configuration, go to the Data Layer item *endpoint-configuration*.

The following configuration parameters (Data Layer items) are available:

Data Layer item	Data type	Description	Default value
<i>allow-deprecated-policies</i>	Bool	Allowing the use of outdated security policies for the secure channel and the user token	false
<i>bind-port</i>	UInt32	Port number (valid range 1 - 65535)	4840
<i>sec-configs[n]</i>	sechan-config	Array of security configurations (endpoints as specified for OPC UA)	sec-configs[0] sec-configs[1] sec-configs[2] sec-configs[3]
<i>sec-configs[n].mode-None</i>	Bool	"true" if the message security mode "none" is permitted	[0] => false [1] => false [2] => false [3] => false
<i>sec-configs[n].mode-Sign</i>	Bool	"true" if the message security mode "sign" is permitted	[0] => false [1] => true [2] => true [3] => true
<i>sec-configs[n].mode-SignAndEncrypt</i>	Bool	"true" if the message security modes "sign" and "encrypt" are permitted	[0] => false [1] => true [2] => true [3] => true
<i>sec-configs[n].policyId</i>	UInt32 (enum SecurityPolicy)	Identifier of the policy used	[0] => 0 [1] => 3 [2] => 4 [3] => 5
<i>sec-policies</i>	Byte array	Index array of the security policies for this endpoint	0,1,2,3

Data Layer item	Data type	Description	Default value
<i>user-token-configs[n]</i>	user-token	Array of user identity token	user-token-configs[0] user-token-configs[1] user-token-configs[2]
<i>user-token-configs[n].policyId</i>	UInt32 (enum SecurityPolicy)	Identifier of the policy used	[0] => 0 [1] => 3 [2] => 5
<i>user-token-configs[n].type</i>	UInt32 (enum UserIdentityTokenType)	UserIdentityTokenType type	[0] => 0 [1] => 1 [2] => 1
<i>user-tokens</i>	ByteArray	Index array of the user token types for this endpoint	0,1,2

The default settings (refer to the default values in the table) correspond to the ones shown in the "Endpoints" section.



Enumerations are currently not supported in the data range of the Data Layer. Thus, write the respective UInt32 value for the fields "policyId" and "type" to sec-configs and to user-token-configs.



The Endpoint configuration also allows to provide an unencrypted OPC UA endpoint. An unencrypted connection is only recommended for diagnostic purposes. The unencrypted connection should not be used and the respective OPC UA endpoint should be disabled for the general operation.

Enumeration SecurityPolicy

Name	Value (UInt32)	Description
NONE	0	↪ http://opcfoundation.org/UA/SecurityPolicy#None
BASIC128RSA15	1	↪ http://opcfoundation.org/UA/SecurityPolicy#Basic128Rsa15
BASIC256	2	↪ http://opcfoundation.org/UA/SecurityPolicy#Basic256
BASIC256SHA256	3	↪ http://opcfoundation.org/UA/SecurityPolicy#Basic256Sha256
AES128_SHA256_RSAPSS	4	↪ http://opcfoundation.org/UA/SecurityPolicy#Aes128_Sha256_RsaOaep
AES256_SHA256_RSAPSS	5	↪ http://opcfoundation.org/UA/SecurityPolicy#Aes256_Sha256_RsaPss



The SecurityPolicies BASIC128RSA15 and BASIC256 are provided due to compatibility reasons. They are not secure and should thus not be used.

UserIdentityTokenType enumeration

Name	Value (UInt32)	Description
ANONYMOUS	0	No token required
USERNAME	1	A username/password token
CERTIFICATE	2	X.509 v3 certificate token
ISSUEDTOKEN	3	Each token created by a authorization service



The token types "ANONYMOUS", "CERTIFICATE" and "ISSUETOKEN" are only supported by the server for testing purposes.

Setting up a connection with one of the token types "ANONYMOUS", "CERTIFICATE" or "ISSUETOKEN" always results in the error message "Bad_IdentityTokenInvalid".

Method add-sec-config

Use the method add-sec-config to add a new security configuration. The added security configuration is attached at the end of the "sec-configs" array.

Method add-user-token-config

Use the method add-user-token-config to add a new UserIdentity configuration. The added security configuration is attached at the end of the "user-token-configs" array.

The input parameters "policyId" and "type" correspond to the fields of the "user-token-configs" structure.

Method remove-sec-config

The remove-sec-config method removes an element from the array of the security configuration "sec-configs".

The input parameter "index" is the index (UInt32) of the array field to be removed.

Method remove-user-token-config

The remove-user-token-config method removes an element from the array of the UserIdentity configuration "user-token-configs".

The input parameter "index" is the index (UInt32) of the array field to be removed.

5.4.3 Session configuration

For the Session configuration, go to the Data Layer item *session-configuration*.

The following configuration parameters (Data Layer items) are available:

Datalayer item	Data type	Description	Default value	Minimum value	Maximum value
<i>keep-spare-session</i>	Bool	If "true", the oldest disconnected session is deleted when num-sessions is reached	true	-	-
<i>lifetime-check-interval</i>	UInt32	Interval in ms during which the session lifetime is checked	5000	100	5000
<i>lifetime-max</i>	UInt32	Maximum session lifetime in milliseconds	3600000	360000	3600000
<i>lifetime-min</i>	UInt32	Minimum session lifetime in milliseconds	10000	1000	50000
<i>max-references-per-node</i>	UInt32	Limits the maximum number of references returned in a browse request	100	1	5000
<i>max-session-calls</i>	UInt32	Maximum number of service calls (per server; not per session!; related to num_uatcpmsg_ctxts)	250	1	500
<i>num-sessions</i>	UInt32	Maximum number of parallel sessions (per server)	12	1	50

5.4.4 Subscription configuration

For the Subscription configuration, go to the Data Layer item *subscription-configuration*.

The following configuration parameters (Data Layer items) are available:

Datalayer item	Data type	Description	Default value	Minimum value	Maximum value
<i>max-lifetime</i>	UInt32	Maximum interval after which there is the Subscription timeout (milliseconds)	100000	1	500000
<i>max-monitored-item-event-queue-size</i>	UInt32	Maximum supported number of events in a monitored item	100	1	500
<i>max-monitored-item-queue-size</i>	UInt32	Maximum supported values of events in a monitored item	100	1	500
<i>max-notifications-per-publish</i>	UInt32	Sum of the values and results in a publish response	1000	1	5000
<i>max-publishing-interval</i>	UInt32	Maximum supported publishing interval	10000	1	50000
<i>max-sampling-interval</i>	UInt32	Maximum supported sampling interval	10000	1	50000
<i>min-lifetime</i>	UInt32	Minimum interval after which there is the Subscription timeout (milliseconds)	1000	1	2000
<i>min-publishing-interval</i>	UInt32	Minimum supported publishing interval	10	1	2000
<i>min-sampling-interval</i>	UInt32	Minimum supported sampling interval	10	1	2000
<i>num-eventfields</i>	UInt32	Number of eventfields	200	1	500
<i>num-eventnotifiers</i>	UInt32	Number of eventnotifiers and eventsources	20	1	50
<i>num-monitored-items</i>	UInt32	Total number of available monitored items	5000	1	5000
<i>num-monitored-items-per-subscription</i>	UInt32	Maximum number of monitored items in a Subscription	500	1	2000
<i>num-notification-messages-per-session</i>	UInt32	Retransmission Queue, has to be at least twice the num_publishrequests_per_session	100	1	200
<i>num-publishrequests-per-session</i>	UInt32	Maximum number of publishrequests in one session in the waiting queue	10	1	150
<i>num-sessions-with-subscriptions</i>	UInt32	Only this number of sessions can obtain a Subscription	10	1	150
<i>num-subscriptions</i>	UInt32	Total number of available Subscriptions	50	1	150
<i>num-subscriptions-per-session</i>	UInt32	Maximum number of Subscriptions in one session	10	1	150

5.5 Web interface

The ctrlX OPC UA Server app is provided via a web interface to access status and configuration data.

After installing the ctrlX OPC UA Server app, this app integrates itself automatically into the web interface of the control (see Fig. 18).

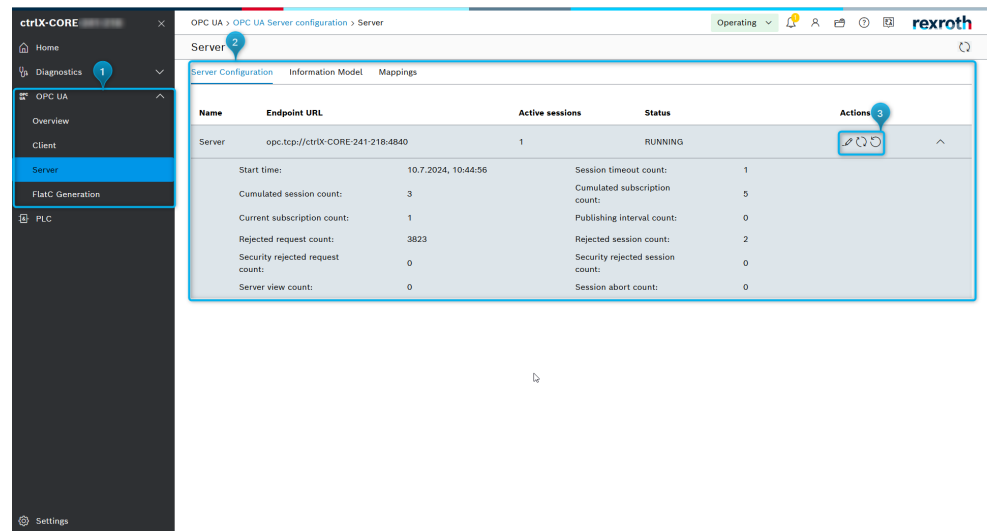


Fig. 18: Web interface of the ctrlX OPC UA Server app

The web interface of the OPC UA app is divided into the following sections:

- ① **side navigation:**
Entry point to the OPC UA-specific settings
- ② **window OPC UA Server configuration:**
Representation of current server data, e.g. the number of current sessions and the server configuration. The tabs “Information Model” and “Mappings” are explained in detail under [Chapter 5.6 OPC UA information models on page 32](#).
- ③ **actions button**
Enables various actions on the ctrlX OPC UA Server.

For basic configurations of the ctrlX OPC UA Server, open the sidebar “Setup Server” using (see Fig. 19).

A changed configuration is applied upon the next app restart.

The server app can be restarted with . This interrupts all current connections.

The server app can be reset and restarted with . This resets all app settings and restarts the server. This interrupts all connections and it might not be possible to reconnect them.

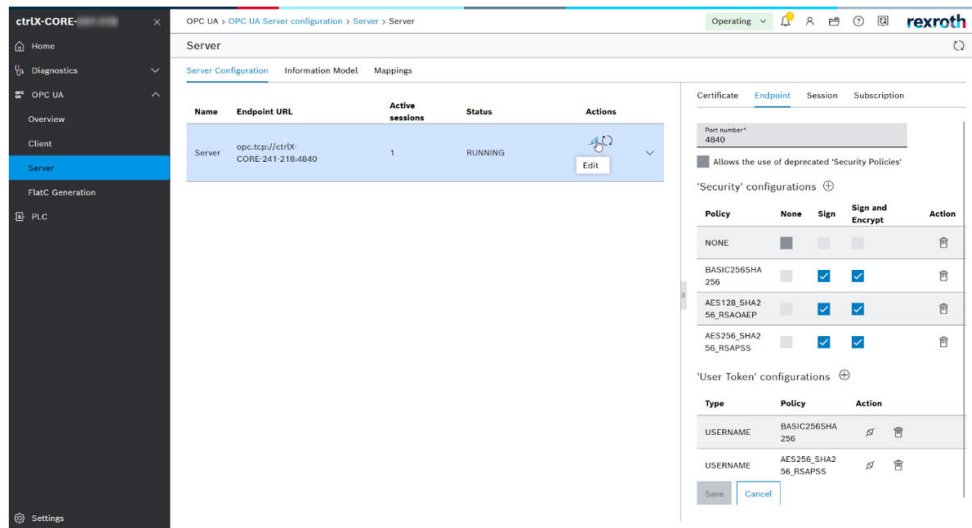


Fig. 19: OPC UA Server - Sidebar when calling 

5.6 OPC UA information models

Loading and modeling of OPC UA information models is a function that enables a user-specific as well as an OPC UA Companion specification-compliant representation of the address space.

In addition, the OPC UA information model nodes (variables and methods) can be mapped to Data Layer data (see [Further information on page 32](#)).

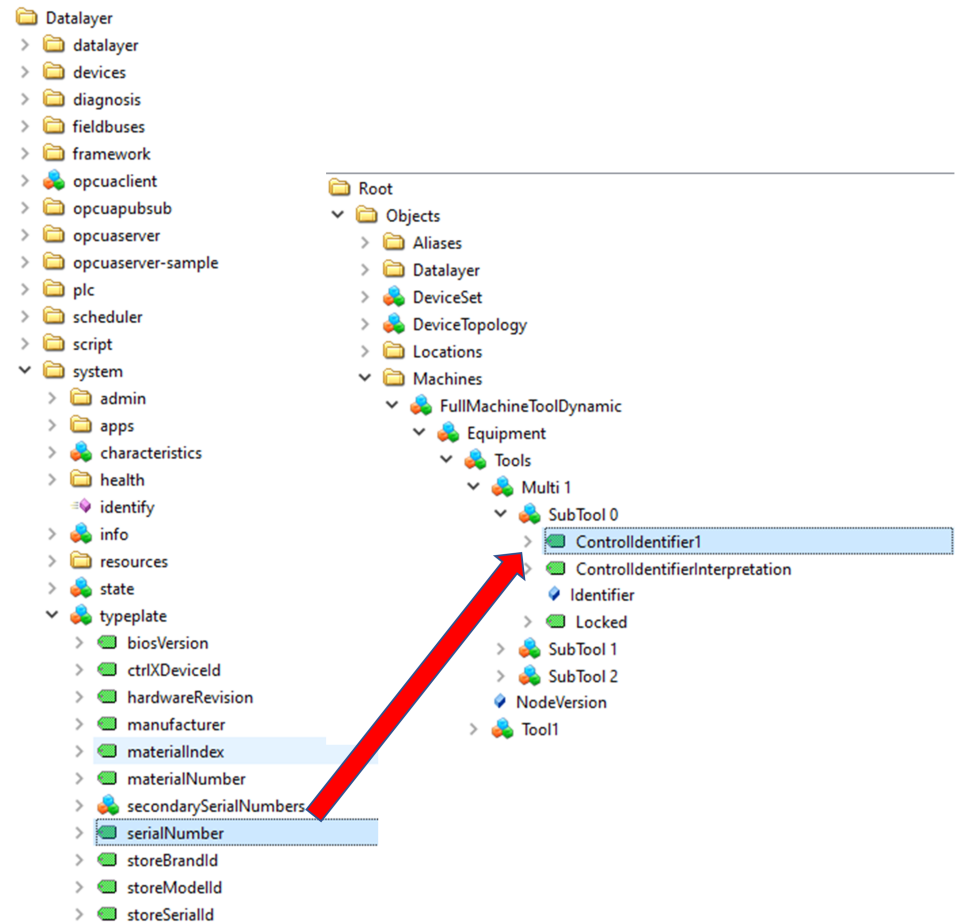


Fig. 20: Mapping Data Layer data to the address space of a loaded OPC UA information model

How To: Loading OPC UA information models

Creating Companion XML model

1. The various Companion specifications provide the OPC UA information models as XML file. These files use the official "NodeSet2.xml" format from the OPC Foundation. The Companion specifications normally contain mainly the type description of the nodes. Depending on the application, corresponding instances have to be created in the XML file.

There are multiple tools to model the address space that use the official "NodeSet2.xml" format.

It can be modeled using the UaModeler from Unified Automation for example:

➔ <https://www.unified-automation.com/products/development-tools/uamodeler.html>

Compiling bin files from models

2. ➤ The "xml2bin" tool can be used to generate binary files from XML models. These files are required to generate the address space in the ctrlX OPC UA Server.

The tool is in the ZIP file "xml2binPackage.zip" in App Data under "*opcua-server/companion-cfg/tools*". The ZIP file "xml2binPackage.zip" contains a "readme.txt" with the respective instructions for execution.

Go to the [ctrlX Developer Community](#) to also find a description of how to use the tool in the article "How-to run an OPC UA Information Model on ctrlX OPC UA Server".

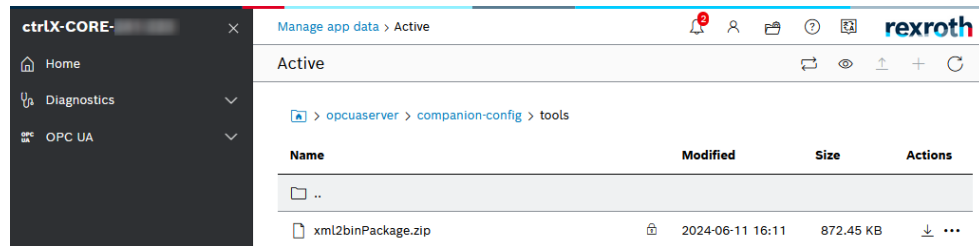


Fig. 21: Storage of "xml2binPackage.zip" in App Data

Loading bin files to the control via "Manage app data"

3. ➤ The generated binary file has to be loaded to the control via the OPC UA Server web interface (refer to the following figure).

The OPC UA Server can also store the files in App Data.

After restarting the OPC UA Server, the address space is loaded and available in the OPC UA Server. The web interface displays the status.

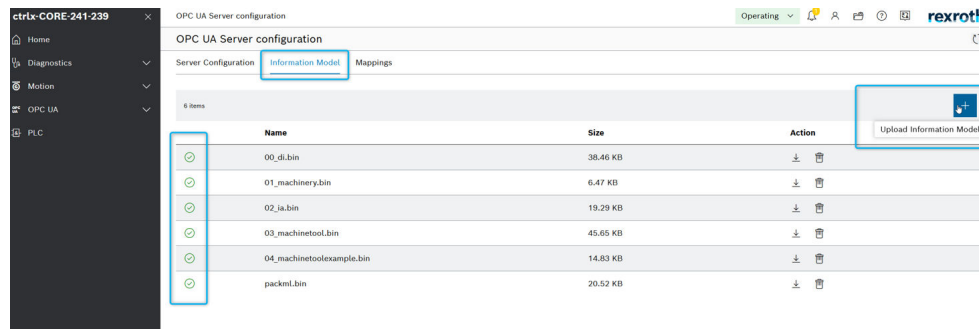


Fig. 22: Storage of the binary files of the OPC UA information models to be loaded

Mapping data to the model

4. In the final step, data can be assigned to the OPC UA information model. The source is Data Layer data. It is mapped via the web interface or the Data Layer node *opcuaserver/companion-config/mappings*.

The following figure shows an exemplary mapping.

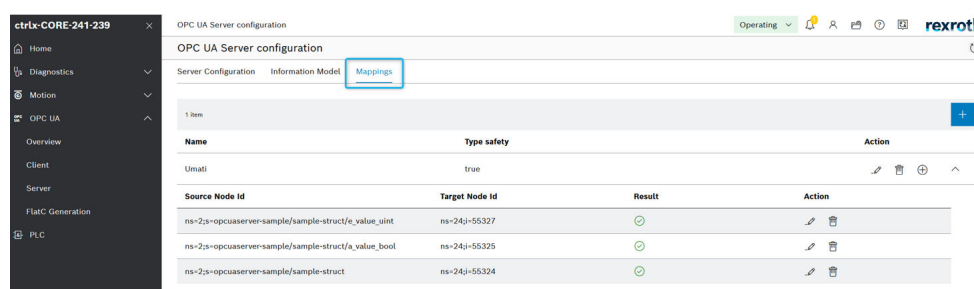
A new mapping can be created by calling "create". This mapping can then be read and written.

It is always mapped via "sourceUaNodeId" and "targetUaNodeId".

"sourceUaNodeId" is the source of data, i.e. the Data Layer node that provides data.

"targetUaNodeId" is the target of the mapping, i.e. the node that is provided by the information model.

The web interface visualizes the "Result", which shows whether it was mapped successfully.



Name	Type safety	Action
Umati	true	[Edit] [Delete] [Add]

Source Node Id	Target Node Id	Result	Action
ns=2;s=opcuaserver-sample/sample-struct/e_value_uint	ns=243i=55327	✓	[Edit] [Delete]
ns=2;s=opcuaserver-sample/sample-struct/a_value_bool	ns=243i=55325	✓	[Edit] [Delete]
ns=2;s=opcuaserver-sample/sample-struct	ns=243i=55324	✓	[Edit] [Delete]

Fig. 23: Mapping configuration in the Data Layer via the web interface of the OPC UA Server

➔ By using the "typeSafety" option, the data type of source and target is compared.

If the OPC UA information model contains structure definitions, these structure definitions can also be used in the Data Layer. After specifying a node - which requires such a data type - in the mapping, this data type is registered in the Data Layer.

An enum definition was created in the following figure.

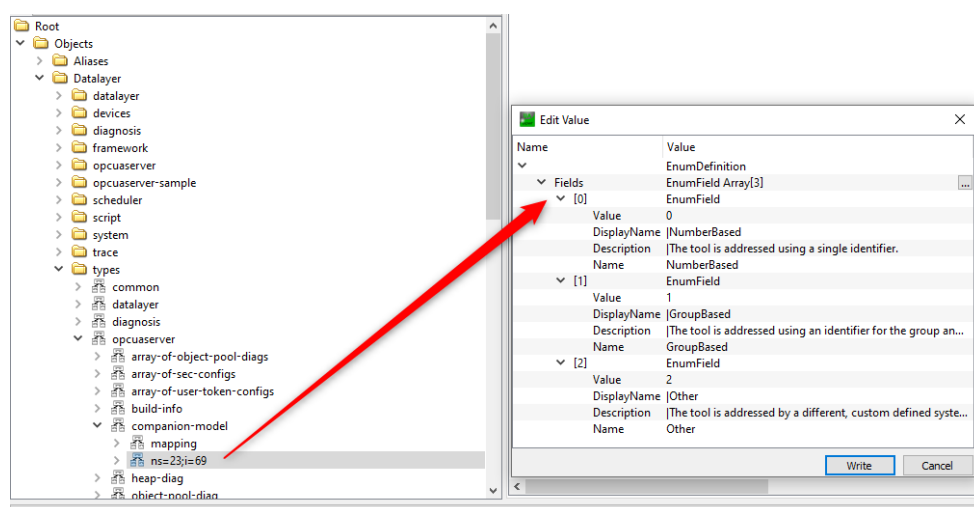


Fig. 24: View on the data types generated for an OPC UA information model in the Data Layer using UaExpert via OPC UA Server access

This FlatBuffer scheme can now be used to define Data Layer data to provide a respective data structure.

6 Related documentation

6.1 Overview

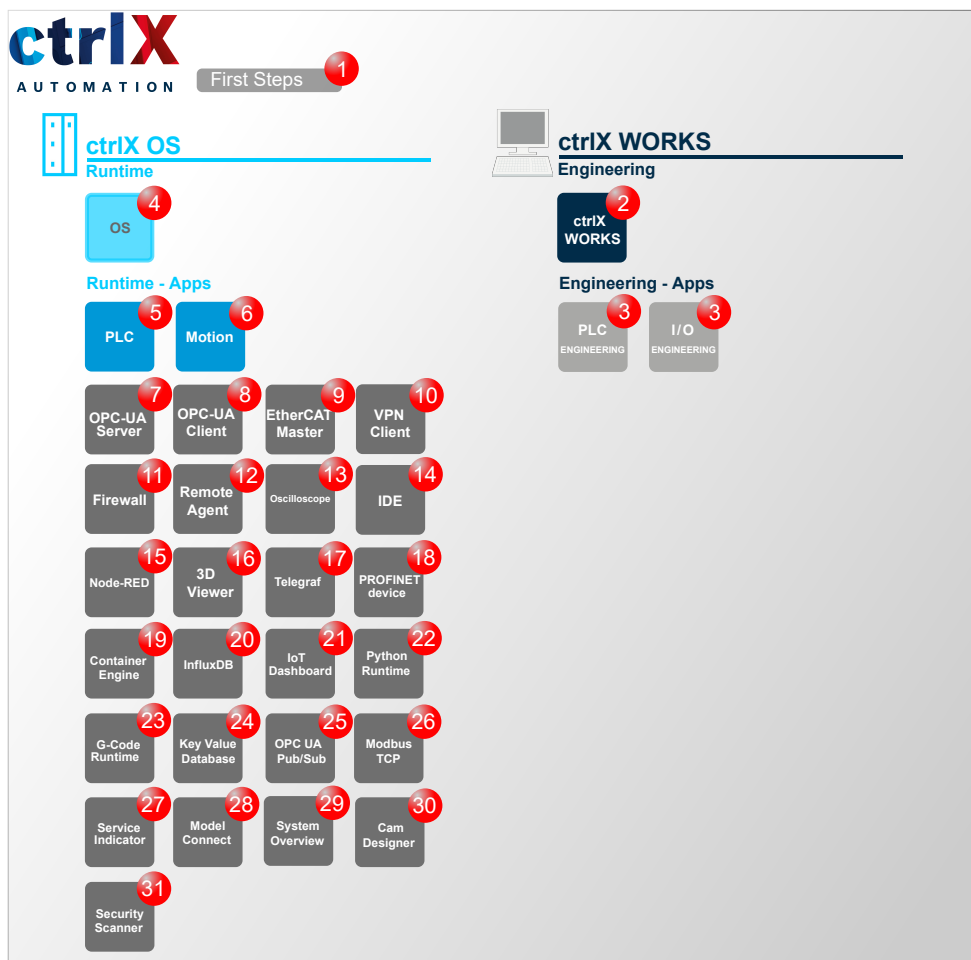


Fig. 25: Overview on further documentations

6.2 ctrlX AUTOMATION

No.	Documentation
1	ctrlX WORKS First Steps Quick Start Guide ↪ Web documentation link Ordering information: • DOK-XWORKS-F*STEP*****-QU01-EN-P • R911403760

6.3 ctrlX WORKS

No.	Documentation
2	ctrlX WORKS - Basic System 03VRS Application Manual ↗ Web documentation link Ordering information: • DOK-XWORKS-WRK***V03**-APRS-EN-P • R911423376
3	ctrlX PLC Engineering - PLC Programming System 03VRS Application Manual ↗ Web documentation link Ordering information: • DOK-XPLC**-PLE***V03**-APRS-EN-P • R911423378
3	ctrlX PLC Engineering - PLC Libraries 03VRS Reference Book ↗ Web documentation link Ordering information: • DOK-XPLC**-LIB***V03**-RERS-EN-P • R911423456

6.4 ctrlX OS

No.	Documentation
4	ctrlX OS - Operating System for ctrlX CORE Control Devices 03VRS Application Manual ↗ Web documentation link Ordering information: • DOK-XCORE*-XCR***V03**-APRS-EN-P • R911423382
	ctrlX OS - Data Layer Nodes 03VRS Reference Book ↗ Web documentation link Ordering information: • DOK-XCORE*-DL****V03**-RERS-EN-P • R911423384
	ctrlX OS - Diagnostics 03VRS Reference Book ↗ Web documentation link Ordering information: • DOK-XCORE*-DIAG**V03**-RERS-EN-P • R911423386

6.5 ctrlX OS apps

No.	Documentation
5	PLC App - PLC Runtime Environment for ctrlX OS 03VRS Application Manual ↪ Web documentation link Ordering information: • DOK-XCORE*-PLC***V03**-APRS-EN-P • R911423401
6	Motion App - Motion Runtime Environment for ctrlX CORE 03VRS Application Manual ↪ Web documentation link Ordering information: • DOK-XCORE*-MOT***V03**-APRS-EN-P • R911423405
7	OPC UA Server App - OPC UA Server for ctrlX OS 03VRS Application Manual ↪ Web documentation link Ordering information: • DOK-XCORE*-UAS***V03**-APRS-EN-P • R911423392
8	OPC UA Client App - OPC UA Client for ctrlX OS 03VRS Application Manual ↪ Web documentation link Ordering information: • DOK-XCORE*-UAC***V03**-APRS-EN-P • R911423390
9	EtherCAT Master App - EtherCAT Master for ctrlX OS 03VRS Application Manual ↪ Web documentation link Ordering information: • DOK-XCORE*-ECM***V03**-APRS-EN-P • R911423394
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